# Towards phase-field modelling of phase inversion

**Thesis submitted in partial fulfilment**

**for the Award of**

INTEGRATED DUAL DEGREE (B.TECH.+ M.TECH.)

**in**

INDUSTRIAL CHEMISTRY

**by**

AASHU SINGH

**METALLURGICAL ENGINEERING**

**INDIAN INSTITUTE OF TECHNOLOGY**

**(BANARAS HINDU UNIVERSITY)**

**VARANASI – 221 005**

ROLL NUMBER
21144027

YEAR OF SUBMISSION
2026

## CERTIFICATE

It is certified that the work contained in the thesis titled **Towards phase-field modelling of phase inversion** by **Aashu Singh** has been carried out under my/our supervision and that this work has not been submitted elsewhere for a degree.

It is further certified that the student has fulfilled all the requirements of Comprehensive Examination, Candidacy and SOTA for the award of **Integrated Dual Degree (B.Tech.+ M.Tech.)**.

**Supervisor**
**Prof. Vikas Jindal**
Metallurgical Engineering
Indian Institute Of Technology
(Banaras Hindu University)
Varanasi – 221 005

## DECLARATION BY THE CANDIDATE

I, **Aashu Singh**, certify that the work embodied in this thesis is my own bona fide work and carried out by me under the supervision of **Prof. Vikas Jindal** from **January 2026** to **May 2026**, at the **Metallurgical Engineering**, Indian Institute of Technology (Banaras Hindu University), Varanasi. The matter embodied in this thesis has not been submitted for the award of any other degree/diploma. I declare that I have faithfully acknowledged and given credits to the research workers wherever their works have been cited in my work in this thesis. I further declare that I have not willfully copied any other's work, paragraphs, text, data, results, *etc.*, reported in journals, books, magazines, reports dissertations, theses, *etc.*, or available at websites and have not included them in this thesis and have not cited as my own work.

Date: May 25, 2026
Place: Varanasi, India

Signature of the Student
Signature

## CERTIFICATE BY THE SUPERVISOR

It is certified that the above statement made by the student is correct to the best of my/our knowledge.

**Supervisor**
**Prof. Vikas Jindal**
Metallurgical Engineering
Indian Institute Of Technology
(Banaras Hindu University)
Varanasi – 221 005

**Signature of Head of Department**

## COPYRIGHT TRANSFER CERTIFICATE

**Title of the Thesis:** Towards phase-field modelling of phase inversion

**Name of the Student:** Aashu Singh

## COPYRIGHT TRANSFER

**The undersigned hereby assigns to the Indian Institute of Technology (Banaras Hindu University), Varanasi all rights under copyright that may exist in and for the above thesis submitted for the award of the Integrated Dual Degree (B.Tech.+ M.Tech.) in Industrial Chemistry.**

Date: May 25, 2026
Place: Varanasi, India

Signature of the Student
Signature

**Note: However, the author may reproduce or authorize others to reproduce material extracted verbatim from the thesis or derivative of the thesis for author's personal use provided that the source and the Institute's copyright notice are indicated.**

# ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my supervisor, **Prof. Vikas Jindal**, for his valuable guidance, continuous support, and encouragement throughout the course of this research work. His knowledge, suggestions, and motivation greatly helped me in completing this thesis successfully.

I would also like to sincerely thank **Dr. Sumanth Nani Enugala** for his constant support, insightful discussions, and valuable suggestions during this work. Although not officially a co-supervisor, his guidance and encouragement contributed significantly to this research.

I am also thankful to the faculty members and staff of the Department of Metallurgical Engineering for providing the necessary facilities and support during my Integrated Dual Degree (B.Tech.+M.Tech.) program.

I would also like to thank my friends and lab members for their help and discussions during this work.

Finally, I am deeply grateful to my family for their constant support, encouragement, and understanding throughout my academic journey.

# Preface

This thesis presents a computational study on precipitate growth and microstructural evolution using phase-field modeling techniques. The work mainly focuses on understanding the effects of elastic interactions and interface curvature on precipitate growth through coupled Allen–Cahn and Cahn–Hilliard equations.

Phase transformations and microstructure evolution play an important role in determining the mechanical and physical properties of materials. In recent years, computational methods have become powerful tools for studying such phenomena. The present work aims to investigate these effects using numerical simulations and elastic energy formulations within the phase-field framework.

The research work presented in this thesis was carried out during my Integrated Dual Degree (B.Tech.+M.Tech.) program in the Department of Metallurgical Engineering. The study provided me with valuable experience in computational modeling, numerical methods, and scientific research.

I hope that the work presented in this thesis contributes toward a better understanding of elastic effects in precipitate evolution and phase transformations.

# Contents

**1 Introduction** **1**

1.1 Introduction . . . . . . . . . . . . . . . . . . . . . . . . . . . . . 1

1.2 Phase-Field Method . . . . . . . . . . . . . . . . . . . . . . . . . . 2

1.3 Elastic Effects in Phase Transformations . . . . . . . . . . . . . . . . . 3

1.4 Numerical Methods for Elastic Field Calculation . . . . . . . . . . . . . 4

1.5 Motivation of the Present Work . . . . . . . . . . . . . . . . . . . . . 5

1.6 Objectives of the Present Work . . . . . . . . . . . . . . . . . . . . . 6

**2 Literature Review** **7**

2.1 Introduction . . . . . . . . . . . . . . . . . . . . . . . . . . . . . 7

2.2 Classical Theory of Precipitate Growth . . . . . . . . . . . . . . . . . . 7

2.3 Development of Phase-Field Modelling . . . . . . . . . . . . . . . . . . 8

2.4 Elastic Effects in Phase-Field Models . . . . . . . . . . . . . . . . . . 8

2.5 Rafting and Elastic Inhomogeneity . . . . . . . . . . . . . . . . . . . 9

2.6 Numerical Methods . . . . . . . . . . . . . . . . . . . . . . . . . . 10

2.7 Research Gap . . . . . . . . . . . . . . . . . . . . . . . . . . . . . 10

**3 Methodology** **12**

3.1 Introduction . . . . . . . . . . . . . . . . . . . . . . . . . . . . . 12

**4 Results and Discussion** **16**

**5 Conclusion** **17**

# List of Figures

# List of Tables

3.1 Parameters used in the simulations. . . . . . . . . . . . . . . . . . . . . . . . 15

# Chapter 1

# Introduction

## 1.1 Introduction

Microstructural evolution plays a crucial role in determining the mechanical, thermal, and physical properties of engineering materials. Processes such as phase separation, precipitate growth, coarsening, and solid state transformations strongly influence material behavior in alloys, ceramics, semiconductors, and functional materials. Understanding these processes is therefore important for the design and optimization of advanced materials systems.

During solid state phase transformations, the evolution of precipitates within a matrix phase is governed by thermodynamic and kinetic factors. In many alloy systems, precipitate growth occurs through diffusion-controlled mechanisms, where atoms diffuse from a supersaturated matrix into the precipitate phase. The morphology, growth rate, and spatial distribution of precipitates significantly affect material properties such as hardness, strength, creep resistance, conductivity, and phase stability.

In addition to chemical free energy, elastic interactions also play an important role in precipitate evolution. When the lattice parameters of the precipitate and matrix phases differ, elastic strain energy develops due to lattice mismatch. This misfit strain alters the local stress field and influences precipitate growth kinetics, morphology evolution, and microstructural stability.

Traditional sharp-interface approaches, such as the Zener–Frank theory and the Laraia–Johnson–Voorhees (LJV) theory, have been widely used to study diffusion-controlled precipitate growth. However, these approaches require explicit interface tracking and become difficult to apply for complex evolving morphologies. To overcome these limitations, diffuse-interface approaches such as the phase-field method have become increasingly important.

## 1.2 Phase-Field Method

The phase-field method is a powerful computational technique used to model microstructural evolution without explicitly tracking interfaces. In this method, interfaces are represented as diffuse regions across which field variables vary continuously. The microstructure is described using continuous order parameters and composition fields.

The major advantage of the phase-field approach is its ability to naturally handle:

- complex interface evolution,
- precipitate coarsening,
- elastic interactions,
- morphology changes,
- and multiphase systems.

The phase-field framework is thermodynamically consistent and is based on the minimization of the total free energy of the system.

For systems involving conserved composition fields, the evolution is governed by the Cahn–Hilliard equation:

$$\frac{\partial c}{\partial t} = \nabla \cdot (M \nabla \mu) \tag{1.1}$$

where:

- $c$ is the composition field,
- $M$ is the mobility,
- $\mu$ is the chemical potential.

For non-conserved order parameters, the evolution is governed by the Allen–Cahn equation:

$$\frac{\partial \eta}{\partial t} = -L \frac{\delta F}{\delta \eta} \tag{1.2}$$

where:

- $\eta$ is the order parameter,
- $L$ is the relaxation coefficient,
- $F$ is the total free energy.

The coupled solution of these equations enables simulation of precipitation, phase separation, and interface evolution.

## 1.3 Elastic Effects in Phase Transformations

Elastic stresses arise during phase transformations because of lattice mismatch between the matrix and precipitate phases. These stresses generate elastic strain energy, which contributes to the total free energy of the system.

The elastic contribution becomes especially important in coherent precipitates, where lattice continuity exists across the interface. The resulting elastic interactions can:

- modify precipitate growth rates,
- influence particle morphology,
- cause directional coarsening,
- and induce rafting phenomena.

The elastic energy is generally expressed as:

$$F^{el} = \frac{1}{2} \int_{\Omega} \sigma_{ij}^{el} \epsilon_{ij}^{el} \, d\Omega \tag{1.3}$$

where:

- $\sigma_{ij}^{el}$ is the elastic stress tensor,
- $\epsilon_{ij}^{el}$ is the elastic strain tensor.

The elastic strain is obtained from:

$$\epsilon_{ij}^{el} = \epsilon_{ij} - \epsilon_{ij}^{0} \tag{1.4}$$

where:

- $\epsilon_{ij}$ is the total strain,
- $\epsilon_{ij}^{0}$ is the eigenstrain or misfit strain.

Mechanical equilibrium in the system is governed by:

$$\frac{\partial \sigma_{ij}}{\partial x_j} = 0 \tag{1.5}$$

Accurate computation of elastic fields is therefore essential for realistic microstructural simulations.

## 1.4 Numerical Methods for Elastic Field Calculation

Several numerical techniques have been developed for solving elastic equilibrium equations in phase-field models. Fourier spectral methods based on Fast Fourier Transform (FFT) are widely used because of their computational efficiency in periodic systems. These methods transform differential equations into algebraic equations in Fourier space and are particularly effective for periodic boundary conditions.

However, FFT-based spectral methods have limitations when non-periodic boundary conditions are required. Many real physical systems involve fixed, open, or non-periodic boundaries where spectral periodicity assumptions are not valid.

To address this limitation, the present work employs the Finite Difference Method (FDM) for solving the governing equations. In the FDM approach, spatial derivatives are approximated using finite difference approximations on discrete grids. This method provides flexibility in handling non-periodic boundary conditions and allows direct implementation of physically realistic boundary constraints.

Unlike FFT-based approaches, the present work solves the elastic equilibrium equations using iterative finite-difference discretization techniques under non-periodic boundary conditions.

## 1.5 Motivation of the Present Work

Most previous phase-field studies of precipitate growth involving elastic interactions have primarily used Fourier spectral methods with periodic boundary conditions. Although computationally efficient, these approaches may introduce artificial periodic interactions and may not accurately represent finite systems or systems with realistic physical boundaries.

The present work is motivated by the need to study precipitate growth using:

- non-periodic boundary conditions,
- finite difference discretization,
- and direct numerical solution of elastic equilibrium equations.

The implementation of finite difference techniques enables greater flexibility in modeling systems where periodicity assumptions are not appropriate.

In addition, the coupled solution of Allen–Cahn and Cahn–Hilliard equations with elastic energy contributions provides a comprehensive framework for studying diffusion-controlled precipitate evolution.

## 1.6 Objectives of the Present Work

The main objectives of the present thesis are:

1. To develop a phase-field model for precipitate growth in elastically stressed systems.
2. To couple the Allen–Cahn and Cahn–Hilliard equations for microstructural evolution.
3. To incorporate elastic strain energy arising from lattice mismatch between precipitate and matrix phases.
4. To solve the elastic equilibrium equations using the Finite Difference Method (FDM).
5. To implement non-periodic boundary conditions in the numerical simulations.
6. To investigate the effect of elastic interactions on precipitate growth and morphology evolution.
7. To compare the simulated microstructural evolution with classical theoretical predictions and previous computational studies.

# Chapter 2

# Literature Review

## 2.1 Introduction

Microstructural evolution in multiphase systems has been extensively studied because of its influence on material properties. Earlier studies mainly focused on precipitate growth, coarsening, and elastic interactions during phase transformations. With the development of computational techniques, phase-field modelling became an important tool for studying such phenomena.

## 2.2 Classical Theory of Precipitate Growth

Zener and Frank developed the classical diffusion-controlled theory for precipitate growth in supersaturated alloys. Their model established the parabolic growth behavior of precipitates under diffusion-controlled conditions.

Later, Laraia, Johnson, and Voorhees incorporated coherent elastic strain energy into precipitate growth theory. Their work showed that elastic strain energy modifies precipitate growth kinetics and equilibrium conditions.

Although these theories provided important understanding of precipitate growth, they were based on sharp-interface assumptions and became difficult to apply for complex evolving microstructures.

## 2.3 Development of Phase-Field Modelling

The phase-field method was developed to study microstructural evolution without explicit interface tracking.

Cahn and Hilliard introduced a diffuse-interface model for conserved composition evolution through the Cahn–Hilliard equation:

$$\frac{\partial c}{\partial t} = \nabla \cdot (M \nabla \mu) \tag{2.1}$$

Allen and Cahn later introduced an evolution equation for non-conserved order parameters:

$$\frac{\partial \eta}{\partial t} = -L \frac{\delta F}{\delta \eta} \tag{2.2}$$

These equations became the foundation of modern phase-field modelling and have been widely used for studying:

- phase separation,
- precipitate growth,
- coarsening,
- and morphology evolution.

## 2.4 Elastic Effects in Phase-Field Models

Khachaturyan developed the microelasticity theory for incorporating elastic interactions into heterogeneous systems. This framework enabled the inclusion of elastic strain energy in phase-field simulations.

Subsequent studies demonstrated that elastic interactions strongly influence:

- precipitate morphology,
- particle alignment,
- directional coarsening,
- and microstructural stability.

Elastic energy in coherent systems is generally expressed as:

$$F^{el} = \frac{1}{2} \int_{\Omega} \sigma_{ij}^{el} \epsilon_{ij}^{el} d\Omega \tag{2.3}$$

The associated mechanical equilibrium condition is:

$$\frac{\partial \sigma_{ij}}{\partial x_j} = 0 \tag{2.4}$$

## 2.5 Rafting and Elastic Inhomogeneity

Several studies investigated rafting behavior in elastically stressed systems.

Schmidt and Gross studied the effect of elastic modulus mismatch and applied stress on precipitate rafting. Their work showed that the direction of rafting depends on:

- **Applied stress**,
- **Eigenstrain**,
- and **Elastic Inhomogeneity**.

Gururajan and Abinandanan developed a phase-field model for studying rafting in elastically inhomogeneous systems. Their simulations demonstrated that elastic modulus mismatch significantly affects precipitate morphology and directional coarsening.

These studies suggested that elastic interactions may also influence topology evolution and phase connectivity.

## 2.6 Numerical Methods

Different numerical approaches have been used in phase-field simulations, including:

- **Finite Difference Method (FDM)**,
- **Finite Element Method (FEM)**,
- **Fourier spectral methods**.

FFT-based spectral methods are widely used because of their computational efficiency and spectral accuracy in periodic systems. However, these methods inherently assume periodic boundary conditions.

Finite Difference Method provides greater flexibility for:

- **non-periodic boundary conditions**,
- **finite-domain simulations**,
- **Iterative numerical implementation**.

Because of these advantages, FDM is suitable for studying topology evolution and connectivity changes in finite systems.

## 2.7 Research Gap

Most previous studies focused primarily on:

- precipitate growth,
- elastic coarsening,
- and rafting phenomena.

Relatively limited work has been carried out on phase inversion using phase-field methods with elastic interactions.

Furthermore, most elastic phase-field studies employed FFT-based spectral techniques together with periodic boundary conditions.

The present work attempts to study topology evolution towards phase inversion using:

- coupled Allen–Cahn and Cahn-Hilliard equations,
- elastic free energy formulation,
- Finite Difference Method,
- and non-periodic boundary conditions.

# Chapter 3

# Methodology

## 3.1 Introduction

The microstructural evolution in our system is governed by the Cahn–Hilliard equation [3.1] and the Allen–Cahn equation [3.2]:

$$\frac{\partial c}{\partial t} = \nabla \cdot (M \nabla \mu) \tag{3.1}$$

$$\frac{\partial \eta}{\partial t} = -L \frac{\delta F}{\delta \eta}, \tag{3.2}$$

where $M$ is the atomic mobility, $\mu$ is the chemical potential, and $L$ is the relaxation coefficient for the order parameter.

Chemical potential $\mu$ is defined as the variational derivative of the total free energy per atom, $F$, with respect to the local composition $c$,

$$\mu = \frac{\delta (F/NV)}{\delta c}. \tag{3.3}$$

The total free energy of the system $F$ is assumed to be given by the sum of a chemical contribution $F^{ch}$ and an elastic contribution $F^{el}$:

$$F = F^{ch} + F^{el}. \tag{3.4}$$

The chemical contribution $F^{ch}$ is given by the following functional:

$$F^{ch} = N_v \int_{\Omega} \left[ f(c,\eta) + K_c(\nabla c)^2 + K_\eta(\nabla \eta)^2 \right] d\Omega, \tag{3.5}$$

where $f(c,\eta)$ is the bulk free energy per atom, $K_c$ and $K_\eta$ are the gradient energy coefficients for gradients in composition and order parameter, respectively, and $NV$ is the number of atoms per unit volume.

The bulk free energy density $f(c,\eta)$ is given by,

$$f(c,\eta) = f^m(c)\left(1 - W(\eta)\right) + f^p(c)W(\eta) + P\eta^2(1-\eta)^2, \tag{3.6}$$

where $P$, a constant, sets the height of the free energy barrier between the m-phase and p-phase.

In Eq. (3.6), $f^m(c)$ and $f^p(c)$ stand for free energies of m (matrix) and p (precipitate) phases, respectively. In our work, they are assumed to take the following simple forms:

$$f^m(c) = Ac^2, \tag{3.7}$$

$$f^p(c) = B(1-c)^2, \tag{3.8}$$

with positive constants $A$ and $B$. In Eq. (3.6), $W(\eta)$ is an interpolation function [15], given by:

$$W(\eta) = \begin{cases} 0, & \text{for } \eta < 0, \\ \eta^3(10 - 15\eta + 6\eta^2), & \text{for } 0 \leq \eta \leq 1, \\ 1, & \text{for } \eta > 1. \end{cases} \tag{3.9}$$

The elastic contribution to the free energy is:

$$F^{el} = \frac{1}{2}\int_X \sigma_{ij}^{el}\epsilon_{ij}^{el}\,dX, \tag{3.10}$$

where

$$\epsilon_{ij}^{el} = \epsilon_{ij} - \epsilon_{ij}^0, \tag{3.11}$$

where $\sigma_{ij}^{el}$ and $\epsilon_{ij}^{el}$ are elastic stress and strain tensors, respectively, and $\epsilon^0$ is the position-dependent eigenstrain (misfit strain). The total strain $\epsilon_{ij}$ is:

$$\epsilon_{ij} = \frac{1}{2}\left(\frac{\partial u_i}{\partial r_j} + \frac{\partial u_j}{\partial r_i}\right), \tag{3.12}$$

where $u_i$ is the displacement field.

Assuming both the m and p phases to be linearly elastic, we have:

$$\sigma_{kl}^{el} = C_{ijkl}\epsilon_{ij}^{el}, \tag{3.13}$$

where $C_{ijkl}$ is the elastic modulus tensor.

The stress field $\sigma_{ij}^{el}$ obeys the equation of mechanical equilibrium:

$$\sigma_{ij,j}^{el} = 0. \tag{3.14}$$

The eigenstrain $\epsilon_{ij}^0$ and the elastic moduli $C_{ijkl}$ are assumed to depend on the order parameter as follows:

$$\epsilon_{ij}^0(\eta) = \beta(\eta)e_T\delta_{ij}, \tag{3.15}$$

$$C_{ijkl}(\eta) = C_{ijkl}^{eff} + \phi(\eta)\Delta C_{ijkl}, \tag{3.16}$$

where $e_T$ is a constant that determines the strength of the eigenstrain, $\delta_{ij}$ is the Kronecker delta, $\beta(\eta)$ and $\phi(\eta)$ are scalar interpolation functions, $C_{ijkl}^{eff}$ is an "effective" modulus, $C_{ijkl}^p$ and $C_{ijkl}^m$ are the elastic moduli tensor of the p and m phases, respectively, and

$\Delta C_{ijkl} = C^p_{ijkl} - C^m_{ijkl}$. All the model parameters used in our simulations are listed in their non-dimensional form in Table 1.

Table 3.1: Parameters used in the simulations.

| **Parameter type** | **Parameter** | **Non-dimensional values** |
|---|---|---|
| Elastic parameters | $G^m$ | 800 |
| | $\nu$ | 0.3 |
| | $\delta$ | 0.5, 1.0 and 2.0 |
| | $e_T$ | 0.01 |
| | $\phi(\eta)$ | $\eta^3(10 - 15\eta + 6\eta^2)$ -1/2 |
| | $\beta(\eta)$ | $\eta^3(10 - 15\eta + 6\eta^2)$ |
| | $C^{eff}_{ijkl}$ | $\frac{1}{2}(C^m_{ijkl} + C^p_{ijkl})$ |
| Simulation parameters | $\Delta x$ | 0.4 |
| | $\Delta y$ | 0.4 |
| | $\Delta t$ | 0.2 |
| | System size | $128 \times 128$ in 2D, $512 \times 512$ in 2D, $1024 \times 1024$ in 2D |
| | Allowed error | $1 \times 10^{-4}$ in displacements |

# Chapter 4

# Results and Discussion

Begin writing your results here.

# Chapter 5

# Conclusion

Begin writing your conclusion here.